

```

#include <stdio.h>
void firstFit(int blockSize[], int m, int processSize[], int n)
{
    int allocation[50];
    int used[50] = {0};
    for(int i = 0; i < n; i++)
        allocation[i] = -1;
    for(int i = 0; i < n; i++)
    {
        for(int j = 0; j < m; j++)
        {
            if(blockSize[j] >= processSize[i] && used[j] == 0)
            {
                allocation[i] = j;
                used[j] = 1;
                break;
            }
        }
    }
    printf("\n--- First Fit ---\n");
    printf("Process No.\tProcess Size\tBlock No.\n");
    for(int i = 0; i < n; i++)
    {
        printf("%d\t%d\t", i + 1, processSize[i]);
        if(allocation[i] != -1)
            printf("%d\n", allocation[i] + 1);
        else
            printf("Not Allocated\n");
    }
}

void bestFit(int blockSize[], int m, int processSize[], int n)
{
    int allocation[50];
    int used[50] = {0};
    for(int i = 0; i < n; i++)
        allocation[i] = -1;
    for(int i = 0; i < n; i++)
    {
        int bestIdx = -1;
        for(int j = 0; j < m; j++)

```

```

    {
        if(blockSize[j] >= processSize[i] && used[j] == 0)
        {
            if(bestIdx == -1 || blockSize[j] < blockSize[bestIdx])
                bestIdx = j;
        }
    }
    if(bestIdx != -1)
    {
        allocation[i] = bestIdx;
        used[bestIdx] = 1;
    }
}
printf("\n--- Best Fit ---\n");
printf("Process No.\tProcess Size\tBlock No.\n");
for(int i = 0; i < n; i++)
{
    printf("%d\t%d\t", i + 1, processSize[i]);
    if(allocation[i] != -1)
        printf("%d\n", allocation[i] + 1);
    else
        printf("Not Allocated\n");
}
}

void worstFit(int blockSize[], int m, int processSize[], int n)
{
    int allocation[50];
    int used[50] = {0};
    for(int i = 0; i < n; i++)
        allocation[i] = -1;
    for(int i = 0; i < n; i++)
    {
        int worstIdx = -1;
        for(int j = 0; j < m; j++)
        {
            if(blockSize[j] >= processSize[i] && used[j] == 0)
            {
                if(worstIdx == -1 || blockSize[j] > blockSize[worstIdx])
                    worstIdx = j;
            }
        }
    }
}

```

```

    }
    if(worstIdx != -1)
    {
        allocation[i] = worstIdx;
        used[worstIdx] = 1;
    }
}
printf("\n--- Worst Fit ---\n");
printf("Process No.\tProcess Size\tBlock No.\n");
for(int i = 0; i < n; i++)
{
    printf("%d\t%d\t", i + 1, processSize[i]);
    if(allocation[i] != -1)
        printf("%d\n", allocation[i] + 1);
    else
        printf("Not Allocated\n");
}
}
int main()
{
    int m, n;
    int blockSize[50], processSize[50];
    printf("1WA24CS218\n");
    printf("Enter number of memory blocks: ");
    scanf("%d", &m);
    printf("Enter sizes of %d memory blocks:\n", m);
    for(int i = 0; i < m; i++)
        scanf("%d", &blockSize[i]);
    printf("Enter number of processes: ");
    scanf("%d", &n);
    printf("Enter sizes of %d processes:\n", n);
    for(int i = 0; i < n; i++)
        scanf("%d", &processSize[i]);
    firstFit(blockSize, m, processSize, n);
    bestFit(blockSize, m, processSize, n);
    worstFit(blockSize, m, processSize, n);
    return 0;
}

```

1WA24CS218

Enter number of memory blocks: 5

Enter sizes of 5 memory blocks:

100 500 200 300 600

Enter number of processes: 4

Enter sizes of 4 processes:

212 417 112 426

--- First Fit ---

Process No.	Process Size	Block No.
1	212	2
2	417	5
3	112	3
4	426	Not Allocated

--- Best Fit ---

Process No.	Process Size	Block No.
1	212	4
2	417	2
3	112	3
4	426	5

--- Worst Fit ---

Process No.	Process Size	Block No.
1	212	5
2	417	2
3	112	4
4	426	Not Allocated